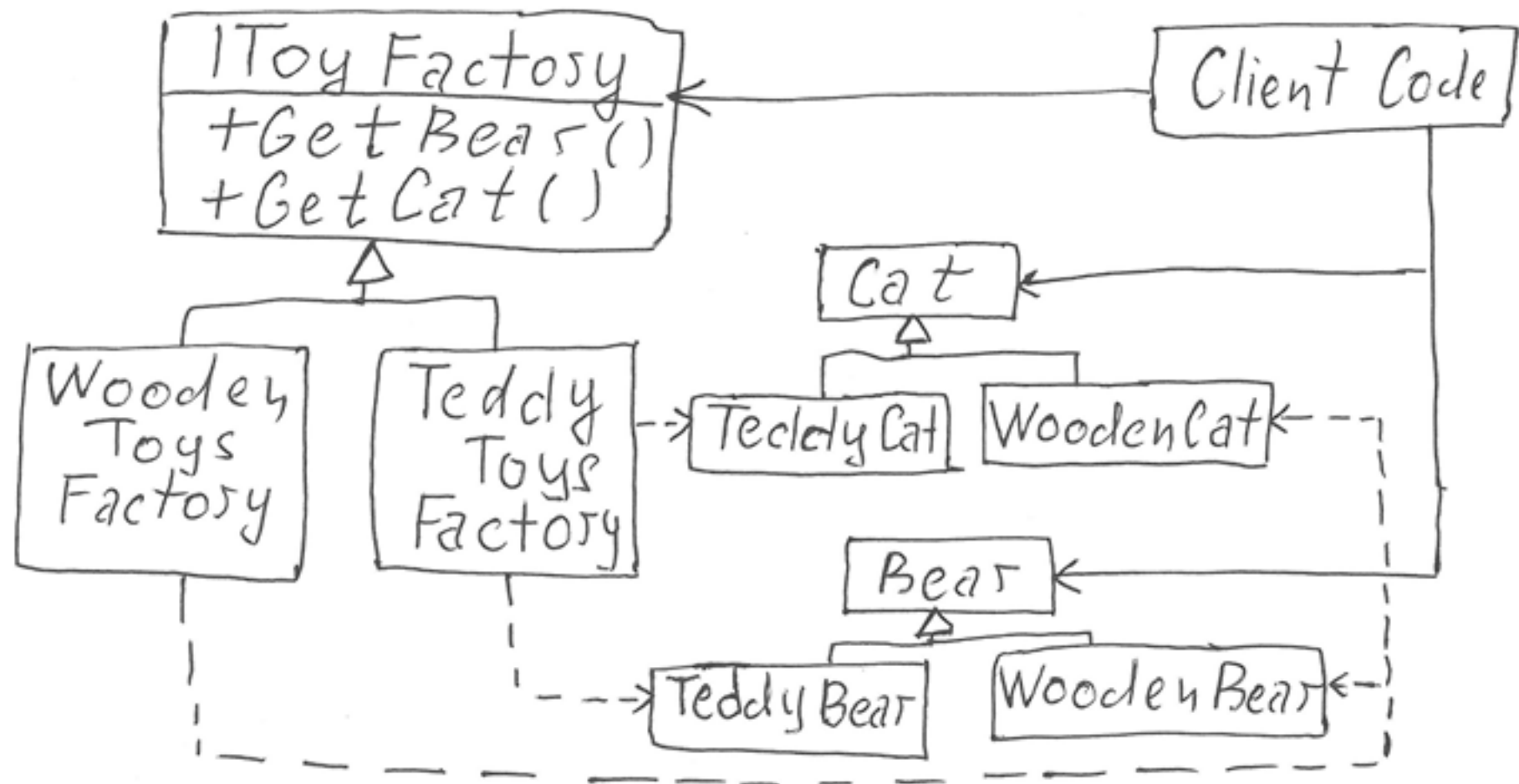
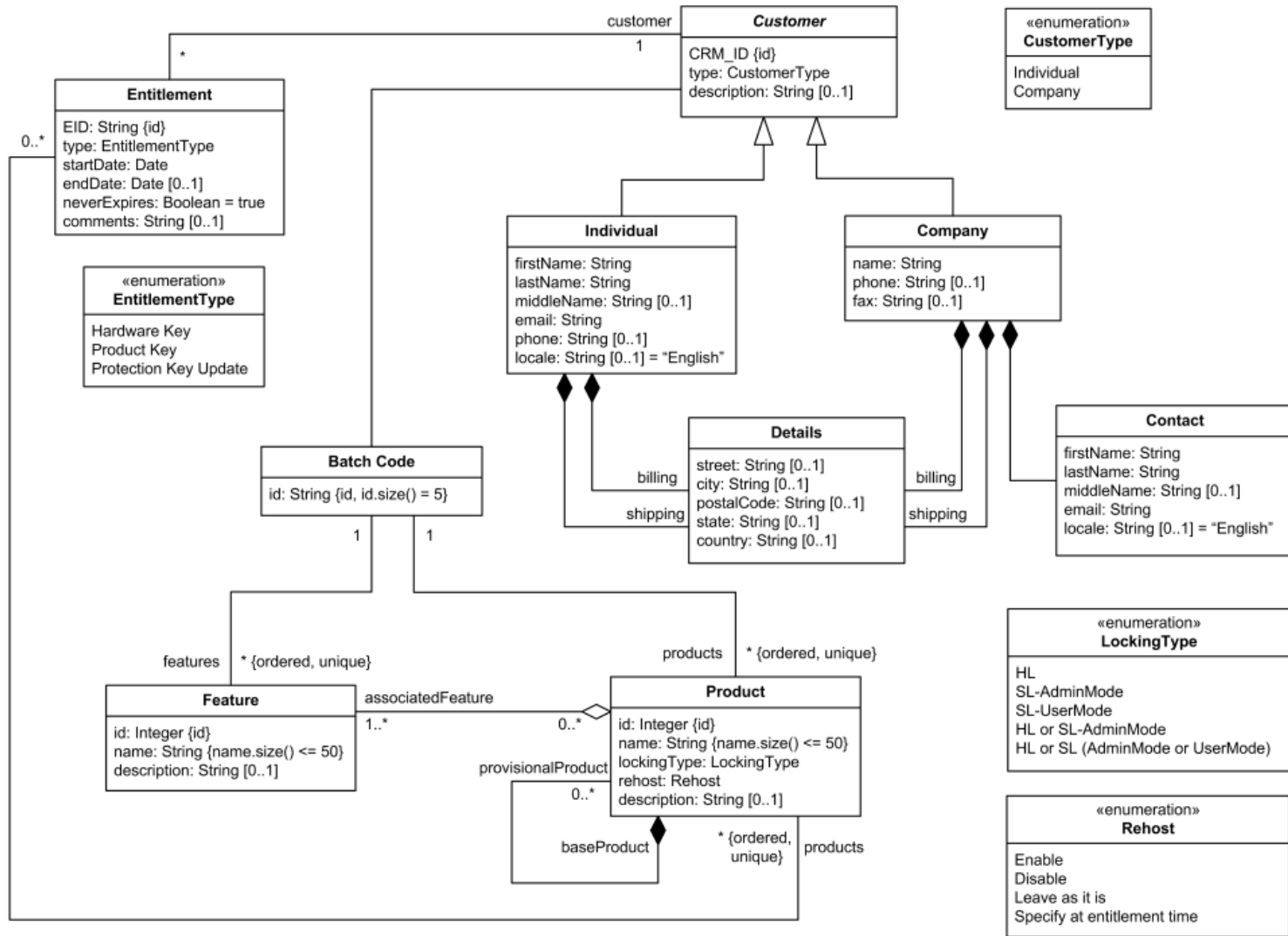


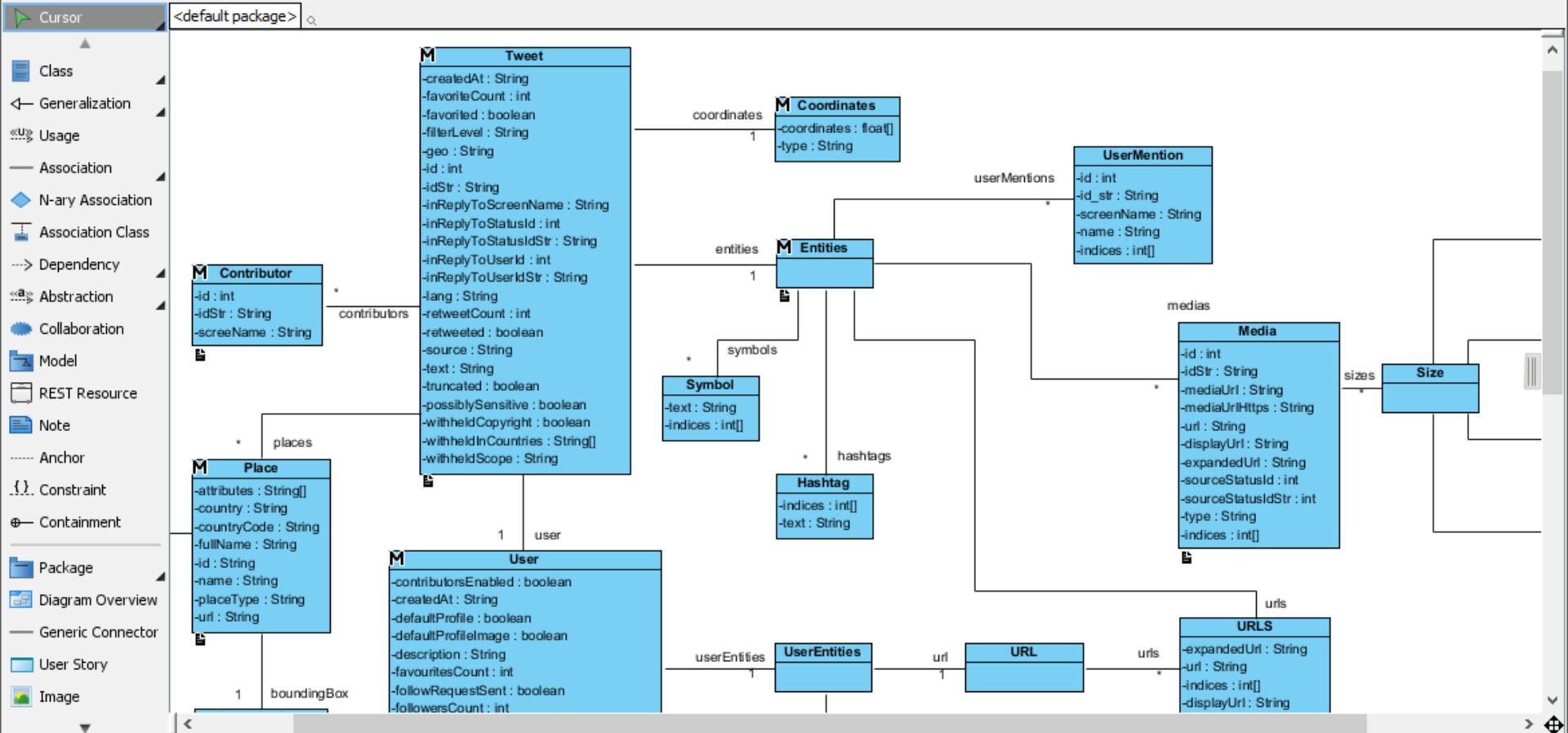
UML System Design Assignment

What you will do

- **UML class diagram for a software system**
- design the structure of the system, **do not implement it**
- UML class diagram **on paper** (A4 or A3)







You are expected to think about

- how objects interact
- how responsibilities are distributed
- how the system can be extended in the future
- how to organise classes so that the design stays clean and understandable

What your diagram must include

- Classes / abstract classes / interfaces
- Hierarchy / composition / aggregation
- meaningful **attributes and methods**
- clear **relationships between classes**

The diagram should clearly represent **how the system works conceptually**
You are not writing Java code, but your design should be **realistic enough that it could be implemented.**

What you should think about

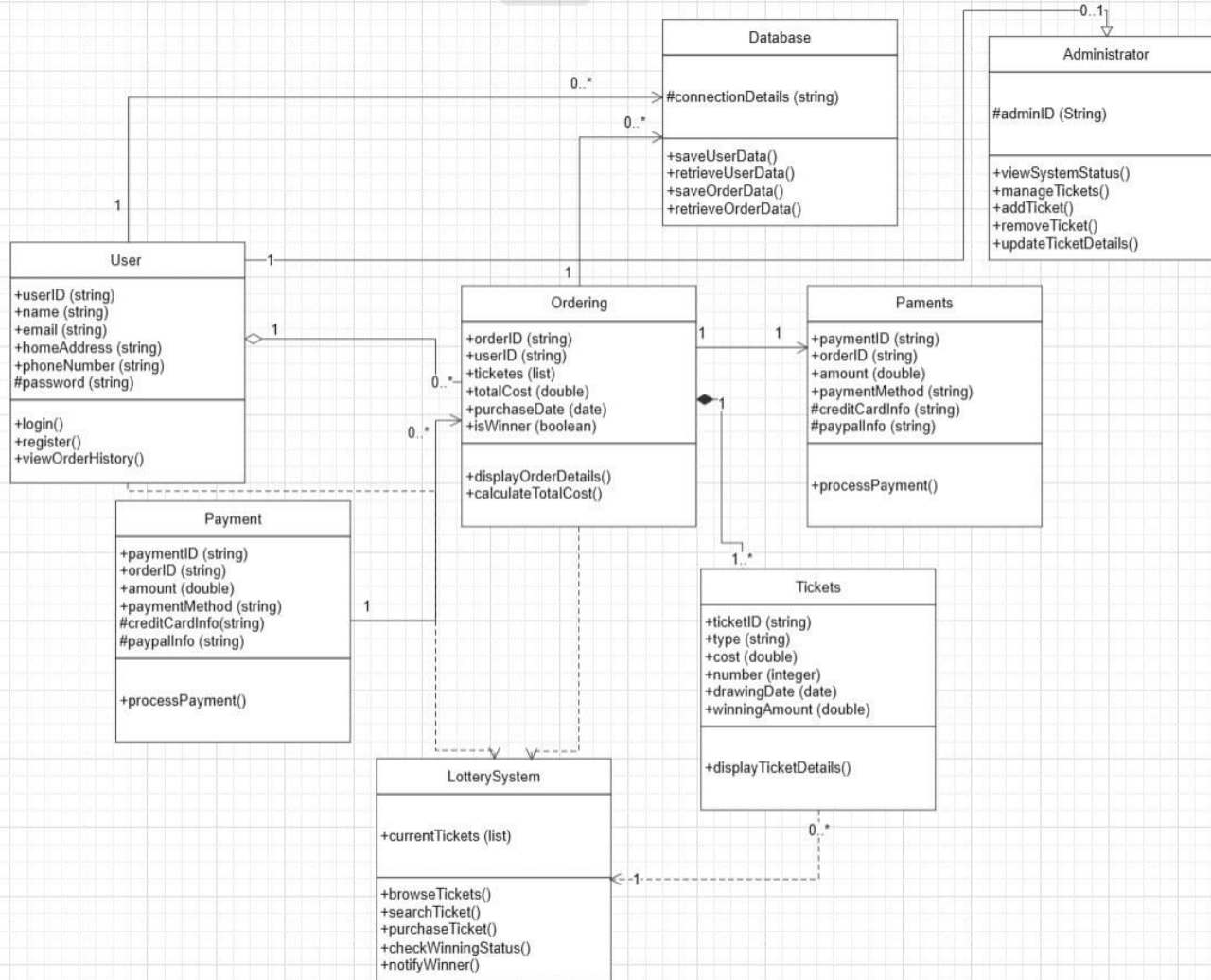
- Which class is responsible for what?
 - Should this be an interface or an abstract class?
 - Should these objects inherit from each other or contain each other?
 - Which class coordinates the system?
 - How will the system handle future extensions?
-
- Avoid creating a **single “god class”** that does everything.
 - Good object-oriented design usually distributes responsibilities across several cooperating classes.

Deliverable

- **one UML class diagram**
- drawn on **A4 or A3 paper**
- clearly labelled
- **readable**

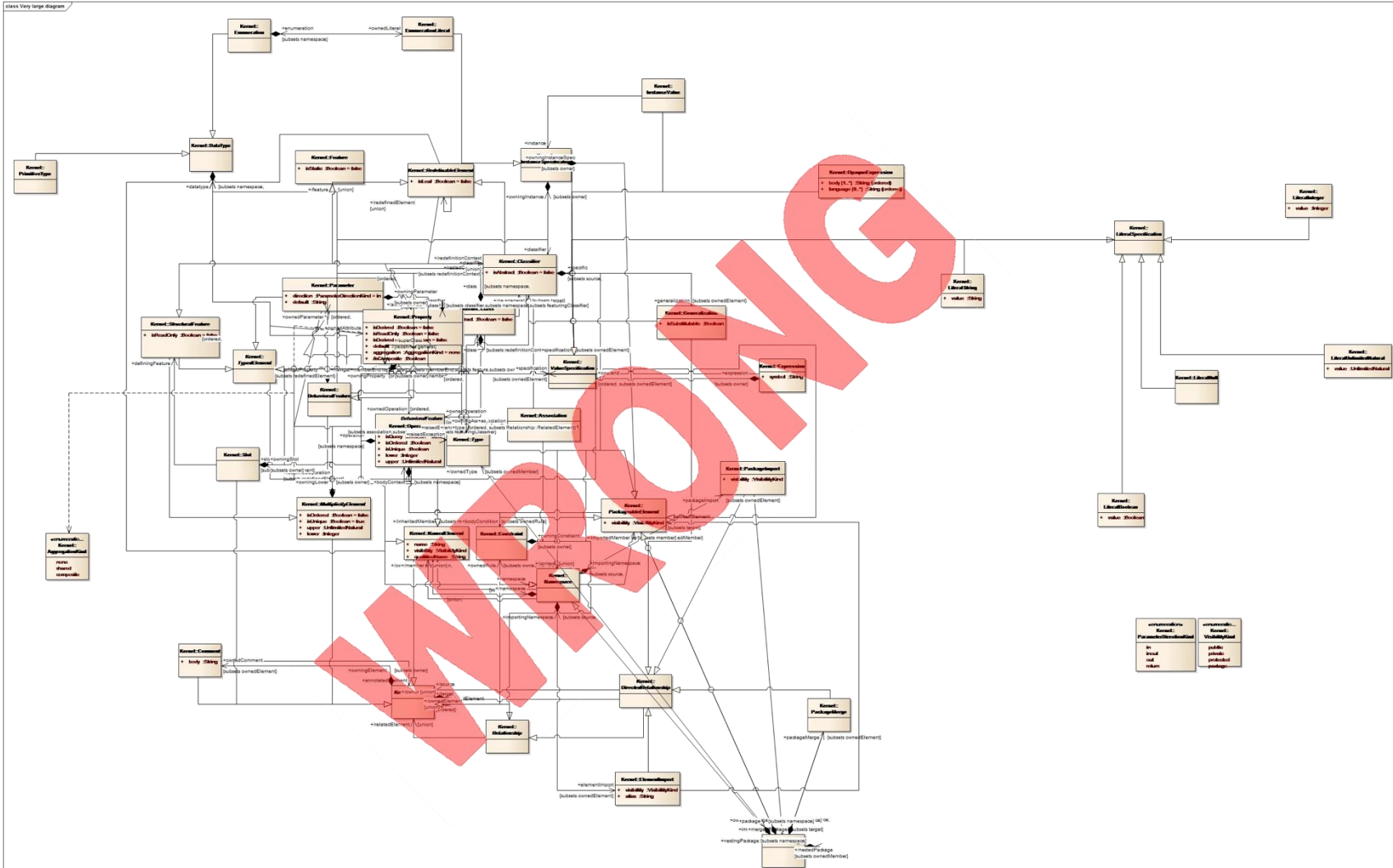
Your diagram should include:

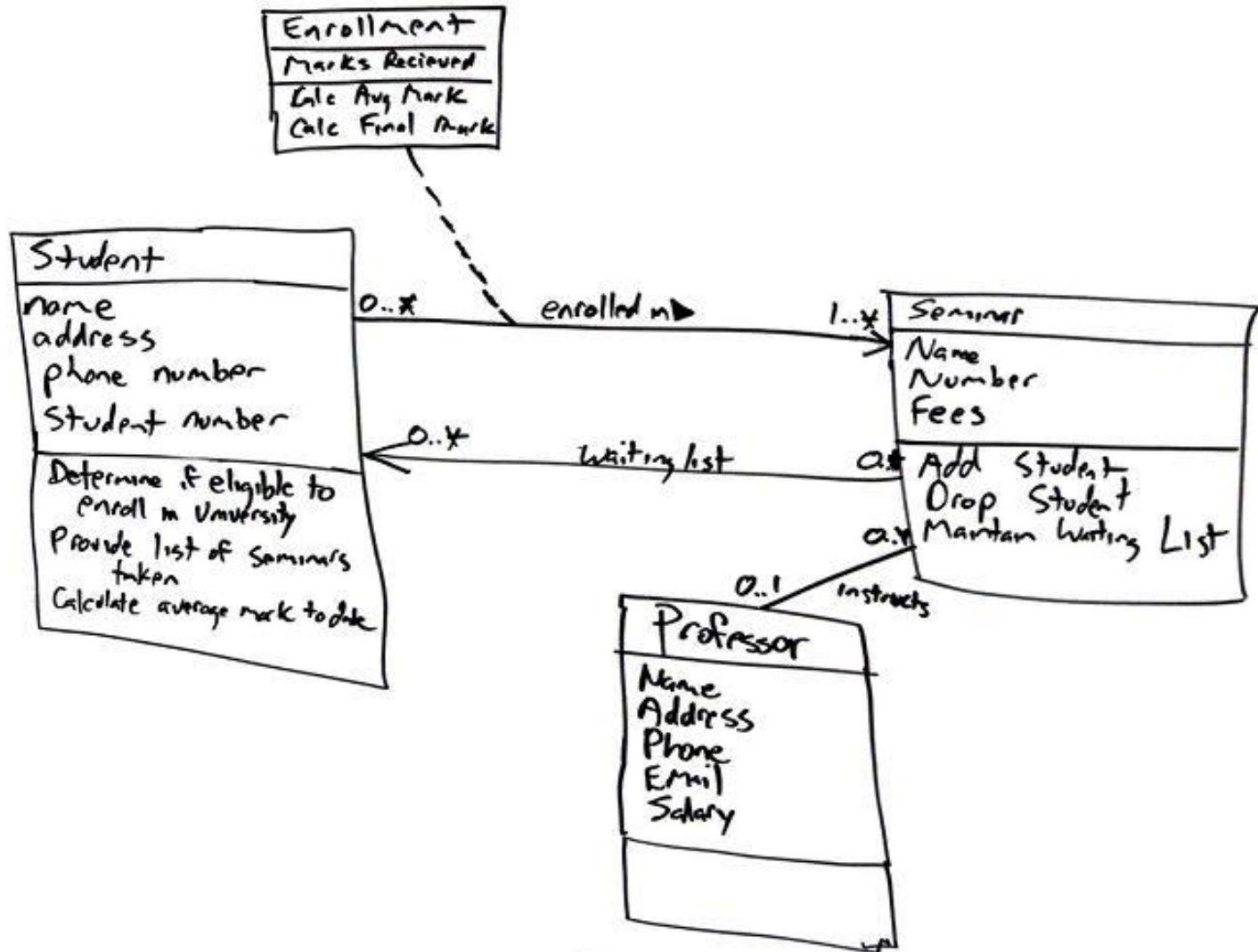
- class names
- important attributes
- key methods
- relationships between classes
- You may optionally add **short notes** explaining design decisions.

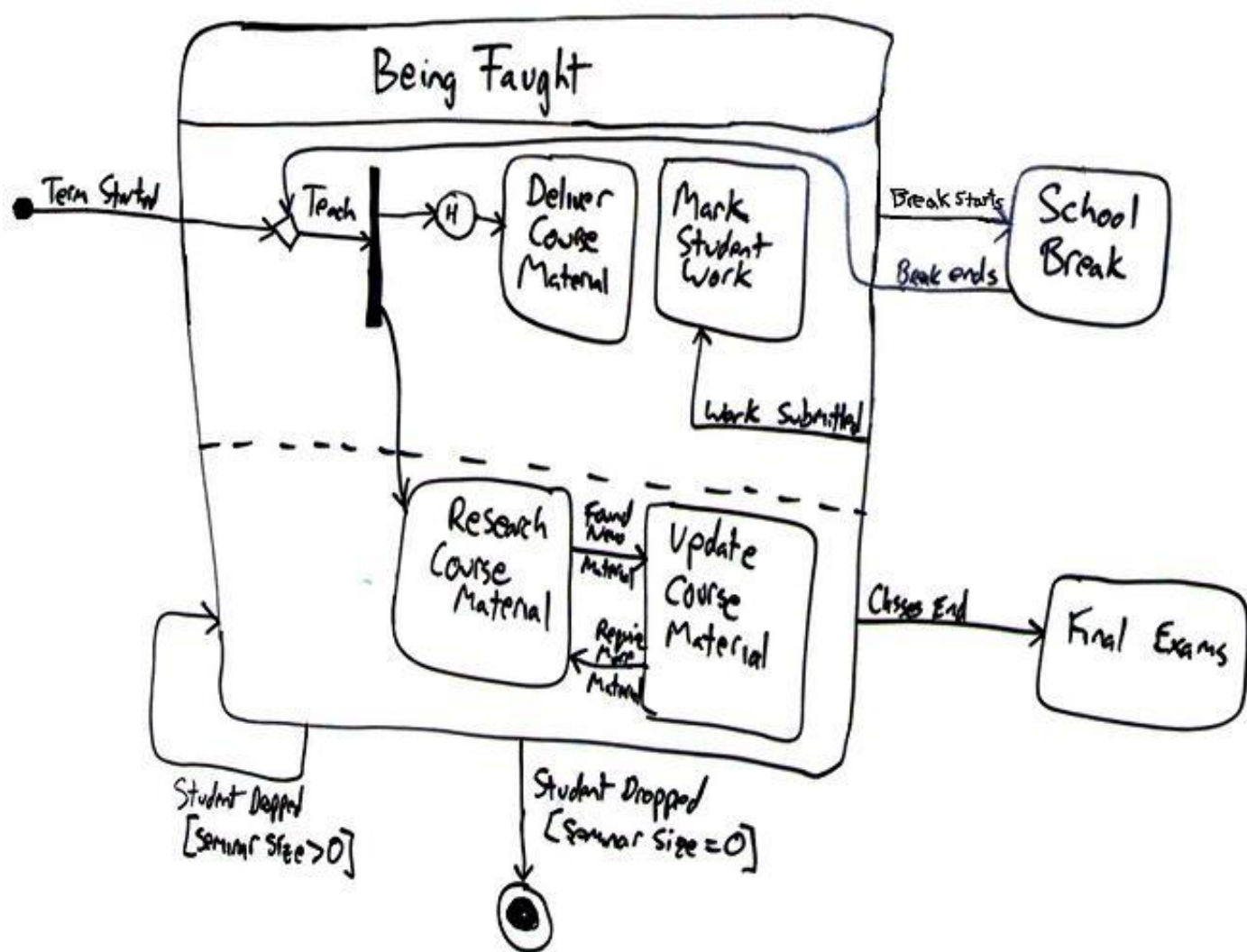


What you should NOT do

- write implementation code
- draw extremely detailed method lists
- invent dozens of unnecessary classes
- create meaningless relationships just to increase class count







Grading Rubric

Total: 15 points

- **System structure 4 points**
 - clear class hierarchy
 - correct use of inheritance
 - meaningful abstraction
- **Object-oriented design 4 points**
 - appropriate separation of responsibilities
 - good use of interfaces or abstract classes
 - avoids overly coupled classes
- **Completeness 4 points**
 - system behaviour can be understood from the diagram
- **Clarity and readability 3 points**
 - diagram is organised
 - labels are readable

Draw.io

UML-class-diagram-tutorial.drawio

File Edit View Arrange Extras Help Unsaved changes. Click here to save. [Share](#)

100%

Misc

Advanced

UML 2.5

UML

Classname

+ field: type

+ field: type

+ field: type

Class 2

Diagram Style

View

☒ Grid 10 pt

☒ Page View

☐ Background [Change](#)

☐ Shadow

Options

☒ Connection Arrows

☒ Connection Points

☒ Guides

Paper Size

A4 (210 mm x 297 mm)

☒ Portrait ☐ Landscape

[Edit Data...](#)

[Clear Default Style](#)

+ More Shapes...

Class diagram tem... Complex class diag... Mermaid class diag... Relationships Page-5

Advice

Start by identifying:

1. the **main objects** in the system
2. the **roles they play**
3. how they **communicate**
4. which objects **own or contain other objects**

Then organise them into a **clean class hierarchy**.

There is **no single correct solution**.